

Audio Spoofing Detection

Model Performance & Diagnostic Evaluation Report

Model Type:	Convolutional Neural Network (CNN)	Date Generated:	June 15, 2026
Framework:	TensorFlow / Keras / Librosa	Evaluation Set:	10 Samples (7 Spoof / 3 Bonafide)

80.00%

ACCURACY

75.00%

F1-SCORE

28.57%

EQUAL ERROR RATE (EER)

1. Executive Summary

This report evaluates the performance of a deep-learning classification network built to distinguish between **Bonafide** (genuine human speech) and **Spoof** (synthetic or replayed) audio signals. The model utilizes Mel-spectrogram feature extractions transformed through sequential 2D convolutional networks. While testing on unseen verification subsets achieves a raw accuracy of 80.00%, the internal metrics reveal significant structural overfitting during training, which warrants architectural revisions before production deployment.

2. Training & Validation Dynamics

The model was trained over a scope of 10 epochs. Optimization and validation indicators outline an acute divergence after the second training epoch:

Phase Subset	Loss Indicator	Accuracy Metric	Behavioral Assessment
Training Set	0.0263	99.13%	Near-perfect adaptation to training distribution features.
Validation Set	8.0507	20.47%	Severe performance degradation indicating strict memorization.

The substantial gap between validation loss escalation and training loss reduction implies that the network optimized parameters toward sample-specific noise rather than generalized structural nuances of audio authenticity.

3. Confusion Matrix Analysis

Testing was conducted on a curated evaluation block consisting of 10 verification signals (7 Spoof, 3 Bonafide). The breakdown of predicted versus empirical designations is structured below:

		Predicted Class	
		Spoof (0)	Bonafide (1)
True Class	Spoof (0)	5 (TN)	2 (FP)
	Bonafide (1)	0 (FN)	3 (TP)

- **True Negatives (TN):** 5 spoof samples accurately isolated and blocked.
- **True Positives (TP):** 3 genuine audio samples successfully verified.
- **False Positives (FP):** 2 fake/spoofed samples mistakenly accepted as real human voice.
- **False Negatives (FN):** 0 genuine users mistakenly rejected.

4. Core Performance Formulations

Based on the matrix values, standard informational retrieval metrics are calculated as follows:

Accuracy

The total ratio of correct classifications across all testing samples.

$$Accuracy = \frac{1}{4} (TP + TN) / Total = (3 + 5) / 10 = 80.00\%$$

Precision

The probability that a sample designated as bonafide is authentically genuine.

$$Precision = \frac{1}{4} TP / (TP + FP) = 3 / (3 + 2) = 60.00\%$$

Recall (Sensitivity)

The capability of the system to discover and authenticate all legitimate users.

$$Recall = \frac{1}{4} TP / (TP + FN) = 3 / (3 + 0) = 100.00\%$$

F1-Score

The harmonic balance between precision and operational recall metrics.

$$F1-Score = 2 \times (Precision \times Recall) / (Precision + Recall) = 75.00\%$$

5. Biometric Verification & Equal Error Rate (EER)

In biometrics and anti-spoofing domains, the **Equal Error Rate (EER)** identifies the optimal operational threshold where the False Acceptance Rate (FAR) equals the False Rejection Rate (FRR).

Arranging the model output probabilities for the evaluation subset shows that at the lower limits, the FRR holds at **0.00%** while the FAR registers at **28.57%** (representing **2/7** spoof elements leaking past constraints). Once the evaluation threshold crosses the baseline genuine confidence mark (**0.98834**), the FRR ascends directly to **33.33%**. Due to this localized step-discontinuity, the formal computed **EER is localized at 28.57%**.

SYSTEM DIAGNOSTIC & ARRESTING OVERFITTING

The model currently demonstrates excellent sensitivity (100% Recall), ensuring zero true users are blocked. However, an EER of 28.57% means almost 3 out of 10 spoof attacks will successfully breach validation. Immediate corrective architectural actions are required.

6. Strategic Optimization Actions

- **Structural Dropout & Regularization:** Implement systematic Dropout layers ($p = 0.4$ to 0.5) following the max-pooling extractions, alongside `L_2` regularizers on weight matrices to force generalized feature mapping.
- **Automated Early Stopping:** Introduce a `tf.keras.callbacks.EarlyStopping` callback tracking validation loss with a patience parameter of 2 epochs to halt execution before model weights lock onto specific training noise patterns.
- **Spectrogram Data Augmentation:** Implement time-stretching, frequency masking, and additive Gaussian acoustic noise elements directly onto training vectors via `librosa` to widen sample variety.